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Dataset	Example (Our Format)	Original Size	Actual Size Used
ParaRel (Elazar et al., 2021)	*Question*: Which country is Georgi Parvanov a citizen of? *Answer*: Bulgaria	*Total data*: 253448	*Training data*: 5575 *ID test data*: 5584 *OOD test data*: 13974
MMLU (Hendrycks et al., 2021)	*Question*: Which of the following did the post-war welfare state of 1948 not aim to provide: (A) free health care and education for all (B) a minimum wage (C) full employment (D) universal welfare. *Answer*: B	*Total data*: 14033	*Training data*: 2448 *ID test data*: 2439 *OOD test data*: 9155
WiCE (Kamoi et al., 2023)	*Evidence*: The first results of the auction for 3DO's franchises and assets... *Claim*: The rights to the M̈ight and Magicñame were purchased for $1.3 million by Ubisoft. *Question*: Does the evidence support the claim? (A) supported (B) partially supported (C) not supported *Answer*: A	*Training data*: 3470 *Dev data*: 949 *Test data*: 958	*Training data*: 3470 *Test data*: 958
HotpotQA (Yang et al., 2018)	*Context*: Arthur's Magazine was an American literary periodical published in ... *Question*: Which magazine was started first Arthur's Magazine or First for Women? *Answer*: Arthur's Magazine	*Training data*: 99564 *Dev data*: 7405 *Test data*: 14810	*Training data*: 10000 *Test data*: 7405
FEVER (Thorne et al., 2018)	*Evidence*: David Bowie is the second studio album by the English musician David Bowie... *Claim*: David Bowie has an album. *Question*: Does the evidence support or refute the claim or not enough information? (A) supports (B) refutes (C) not enough info *Answer*: A	*Training data*: 145449 *Dev data*: 9999 *Test data*: 9999	*Training data*: 10000 *Test data*: 9999
SelfAware (Yin et al., 2023)	*Answerable Question*: What is Nigeria's northernmost climate? *Answer*: rain forest *Unanswerable Question*: Often called high energy particles, what gives life to them? *Answer*: None	*Answerable Question*: 2337 *Unanswerable Question*: 1032	*Unanswerable*: 1032
HaluEval (Li et al., 2023a)	*Knowledge*: Jonathan Stark (born April 3, 1971) is a former... *Question*: Which tennis player won more Grand Slam titles, Henri Leconte or Jonathan Stark? *Answer*: Jonathan Stark	*QA-data*: 10000 *Dialogue*: 10000 *Summarization*: 10000 *User query*:5000	*QA-data*: 10000
FalseQA (Hu et al., 2023)	*Unanswerable Question*: List the reason why mice can catch cats? (This is a question that contradicts common sense)	*Unanswerable Question*: 2365	*Unanswerable*: 2365
NEC (Liu et al., 2024)	*Unanswerable Question*: How long is the typical lifespan of Leogoteo in the wild? (There is no such creature called Leogoteo.)	*Unanswerable Question*: 2078	*Unanswerable*: 2078

Table 6: Illustration and statistics of the datasets. For ParaRel and MMLU, we manually split the datasets into training and test sets. For WiCE, HotpotQA, and FEVER, we directly use the original training set. For SelfAware, FalseQA, and NEC, we directly test models on their unanswerable questions.

A Appendix

A.1 Datasets

We conduct our experiments on nine datasets, which are described as follows.

- **ParaRel** (Elazar et al., 2021): a dataset of factual knowledge with various prompts and relations that are originally for mask prediction. To align the dataset with the requirements of our auto-regressive models, we first change the format into question-answering and our models read the questions and generate the answers. Then, duplicated prompts of different templates but with the same entities are omitted for our question-answering task. It finally comes up with 25, 133 prompt-answer pairs of 31 domains. We split the ParaRel into two sets - the first 15 domains as in-domain data and the last 16 domains as out-of-domain data. We also equally split the in-domain data into training data and test data.
- **MMLU** (Hendrycks et al., 2021): MMLU covers 57 tasks including mathematics, computer science, history, law, and more, which requires extensive world knowledge and problem-solving ability. The dataset is of multiple-choice format, and we can directly use it in our experiments.
- **WiCE** (Kamoi et al., 2023): WiCE is a natural language inference (NLI) dataset for textual entailment. Each data sample consists of evidence and a claim, and the model should decide whether the evidence supports, partially supports, or doesn't support the claim. We turn the dataset into multiple-choice questions with 3 choices for each question.
- **HotpotQA** (Yang et al., 2018): HotpotQA is a question-answering dataset that requires complex reasoning among documents. We evaluate by providing the context documents and questions to see if the model can answer them. Since the test set of HotpotQA requires answer submission, we instead use the development set to do the evaluation.
- **FEVER** (Thorne et al., 2018): FEVER is a dataset containing claims and supporting knowledge. The claims are classified as SUPPORTED, REFUTES, or NOT ENOUGH INFO. We turn it into a multiple-choice NLI task.
- **SelfAware** (Yin et al., 2023): a dataset containing both answerable questions and unanswerable questions. We evaluate the unanswerable questions. It is expected to see our finetuned models refusing the unanswerable questions while other baselines do not possess such ability.
- **HaluEval** (Li et al., 2023a): HaluEval is a dataset containing question-answering, dialogue, summarization, and user-query with correct answers and hallucinated answers. We only take the question-answering part.
- **FalseQA** (Hu et al., 2023): FalseQA is a new open-domain dataset with questions inconsistent with common sense. There are no correct answers to the questions.
- **NEC** (Liu et al., 2024): NEC is also a new open-domain dataset with questions containing some make-up concepts. There are also no correct answers to the questions.

For question-answering tasks, to compare the answer generated by our model with the ground-truth answer, we examine whether the first few output tokens contain the ground-truth answer. We don't adopt exact matching (EM) as the generation is not strictly controllable. For multiple-choice questions, we restrict the model to generate one token and select the choice with maximum probability among the candidate choices by $\text{argmax}_{x \in C}\, logits(x)$, where C is the set of candidate choices. Considering the huge size of HotpotQA and FEVER, we randomly sample 10K training data from them, respectively. More details about the original datasets are shown in Appendix A.1 and Table 6. In Figure 6, we present the distribution of constructed refusal-aware data D_0 and D_1.

Details about the original datasets are shown in Table 6. In Figure 6, we present the distribution of constructed refusal-aware data D_0 and D_1.

A.2 Implementation

We choose OpenLLaMA-3B (Geng and Liu, 2023), LLaMA-7B, and LLaMA-13B (Touvron et al., 2023) as the base models in our experiments. We use LMFlow[3] (Diao et al., 2023a) to conduct instruction tuning, setting epoch to 1, learning rate to $2e^{-5}$, and batch size to 4. All the experiments are implemented on Nvidia A100-40GB GPUs. We conduct experiments with a hyper-parameter sweep consisting of learning rates in $\{1e^{-5}, 2e^{-5}, 5e^{-5}\}$ and batch-size in $\{2, 4, 8\}$ on the training set.

A.3 Label Replacement

In the main experiments, we adopt the padding method for data construction. In addition to

[3] https://github.com/OptimalScale/LMFlow